OzGrav
ARC CENTRE OF EXCELLENCE FOR GRAVITATIONAL WAVE DISCOVERY

# ECHOES OF DYING STARS

## FROM COSMIC BURSTS TO RIPPLES IN SPACETIME

SEA LAKE ASTROFEST
1 NOV 2026

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# APPETIZER



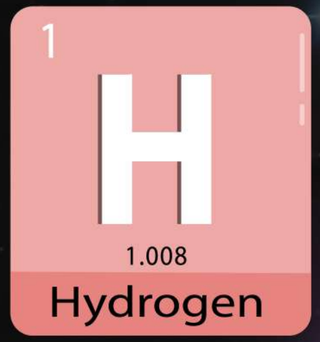
What is a star?

Birth of stars

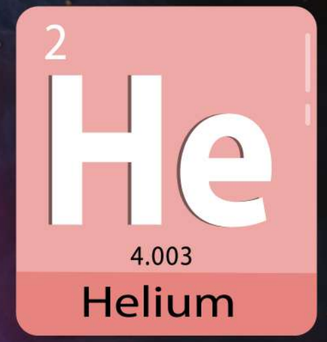
Life of stars

Corpses of massive stars

# STARS IN A NUTSHELL






Gravity

Nuclear fusion

# COLORS OF STARS



Blue



Blue-White



Yellow



Red



Orange

# COLORS OF STARS

Hottest

Blue
~25,000°C

Blue-White
~10,000°C

Yellow
~6,000°C

Orange
~4,000°C

Red
~3,000°C

Coldest

# STAR FORMATION

1

Gas cloud
a.k.a.
Nebula





2

Clumping of
dense gas
pockets

# STAR FORMATION

1

Gas cloud
a.k.a.
Nebula

2

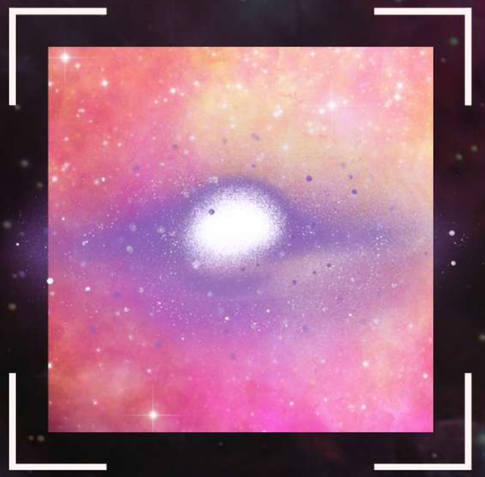
Clumping of
dense gas
pockets

3



Gravitational
collapse

4



Protostar
with
accretion disc

5



# LIFE OF AN AVERAGE STAR



Average star
(like our sun)



Red giant



Planetary nebula



White dwarf

# LIFE OF A MASSIVE STAR



Massive star

more than ~8 times the mass of our sun



Red supergiant



Supernova



Neutron star



Black hole

# LIFE OF A MASSIVE STAR



Massive star

more than ~8 times the mass of our sun



Red supergiant



Supernova





Neutron star



Black hole



Are they really dead?

# MAIN COURSE





What is gravity?

Cosmic ripples a.k.a. Gravitational waves

Hunting cosmic ripples



Afterlife of stars



# FOUNDING FATHERS OF GRAVITY




Early 1600s

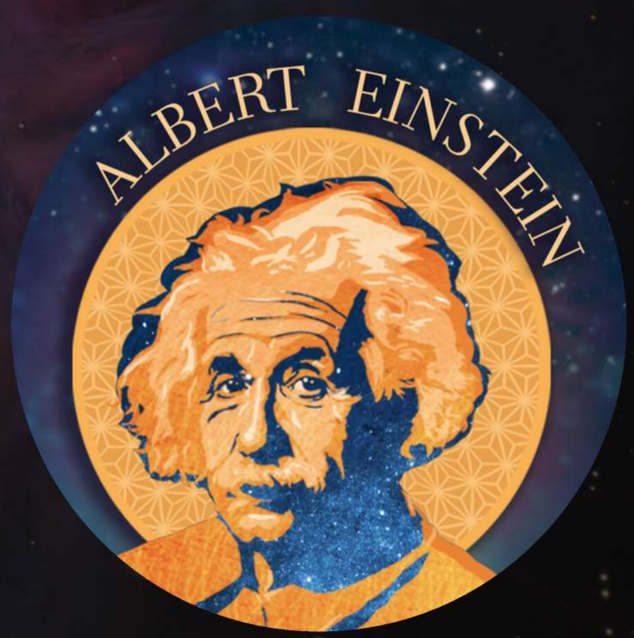






Late 1600s

Early 1900s



# HOW THINGS MOVE AND FALL

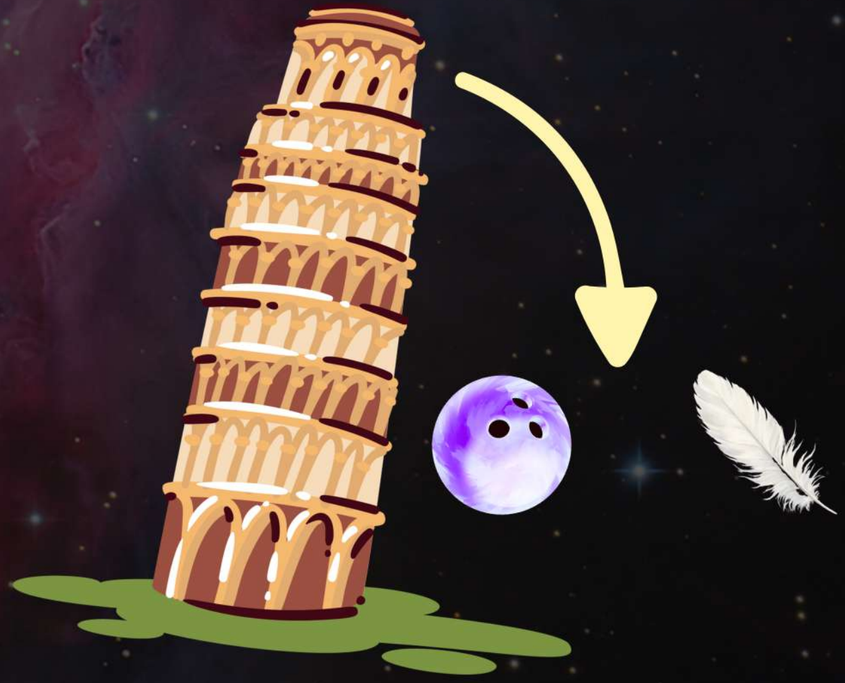





In vacuum,
objects of different masses
fall at the same rate



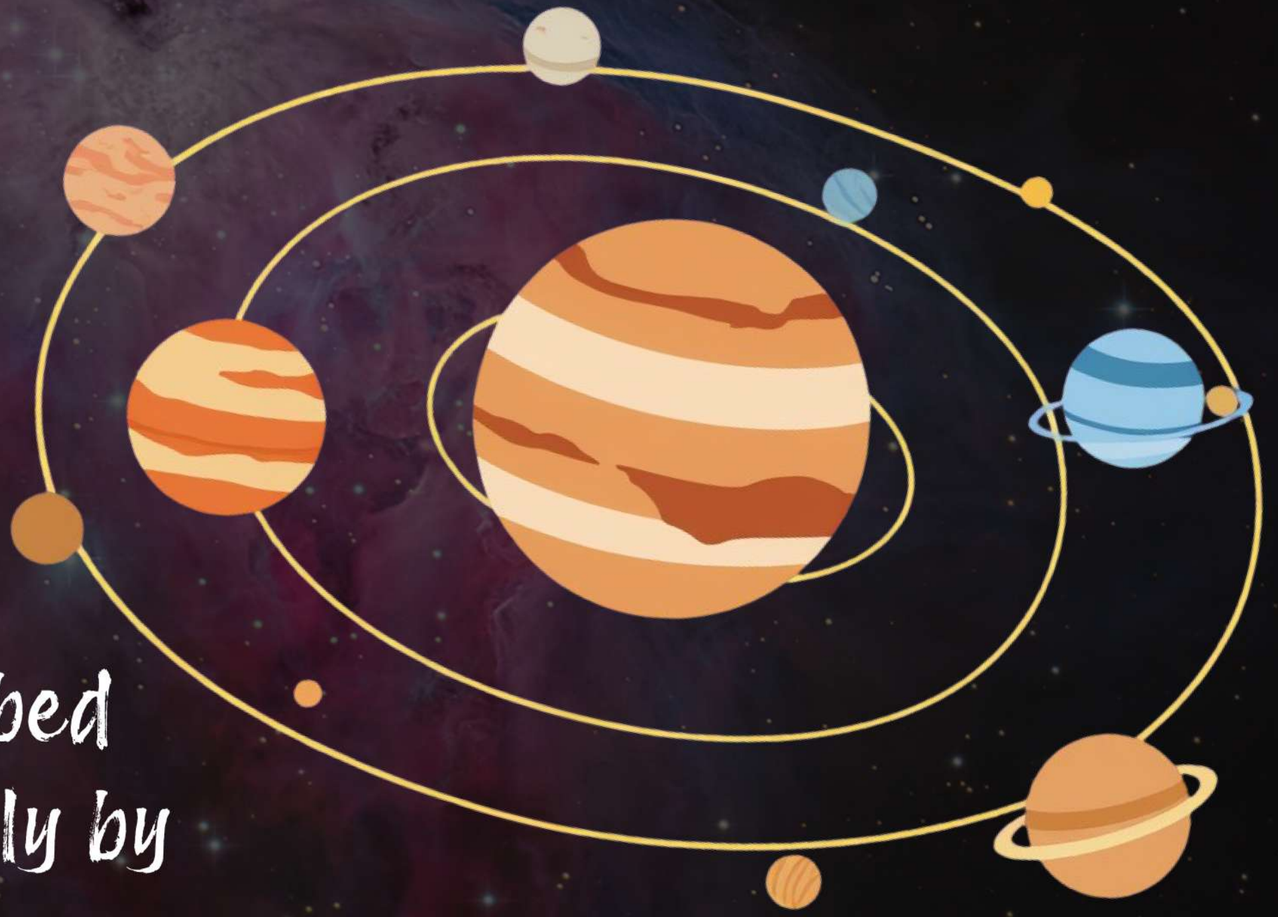
Objects can orbit
celestial bodies
other than Earth.



# MOTION OF PLANETS

Orbital paths follow
predicatble
geometrical shapes



Can be described
mathematically by

THREE LAWS OF PLANETARY MOTION

# UNIVERSAL LAW OF GRAVITY

Unified the motion of objects
on Earth and celestial orbits



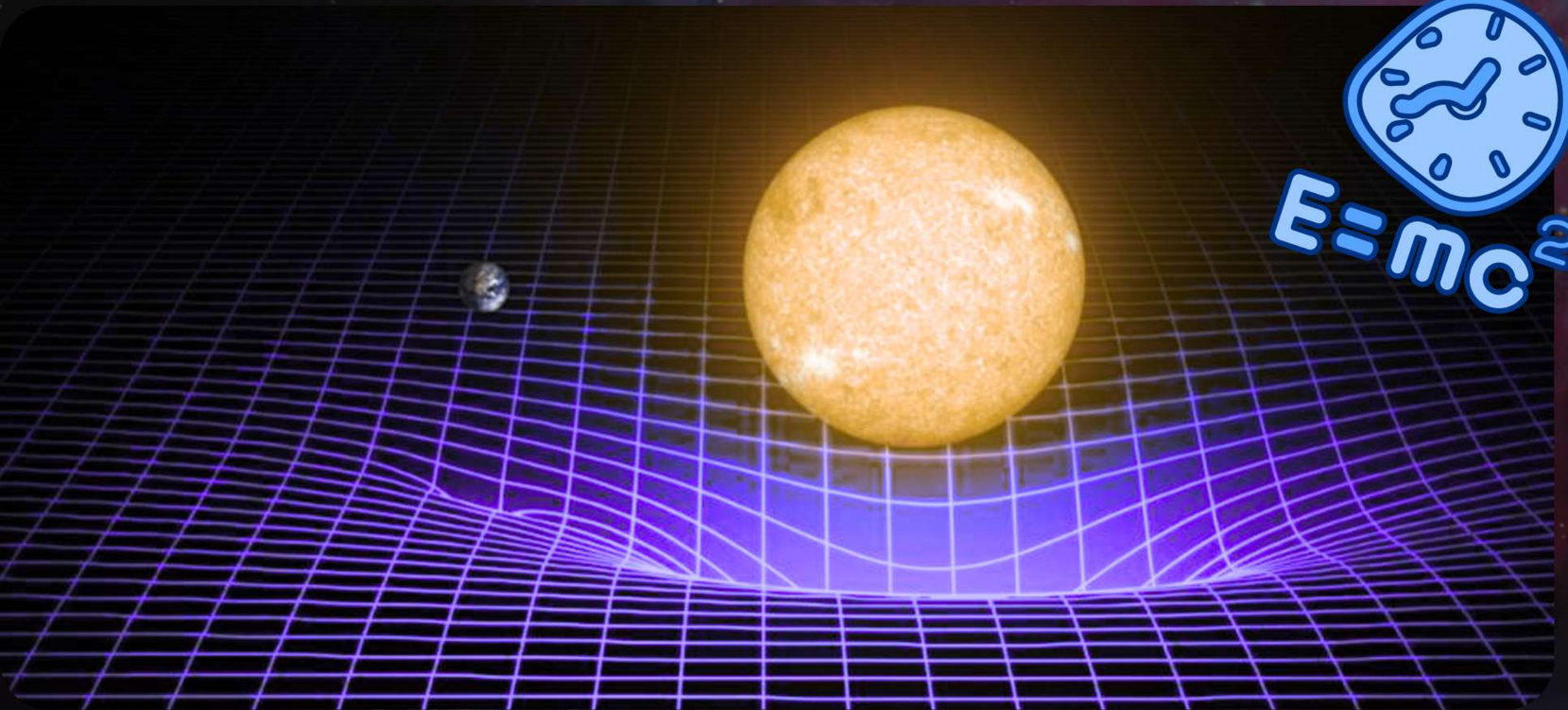



$$F = G\frac{m_1 m_2}{r^2}$$

Objects with mass exerts a force on
every other object with mass.

# BUT WHAT IS GRAVITY?

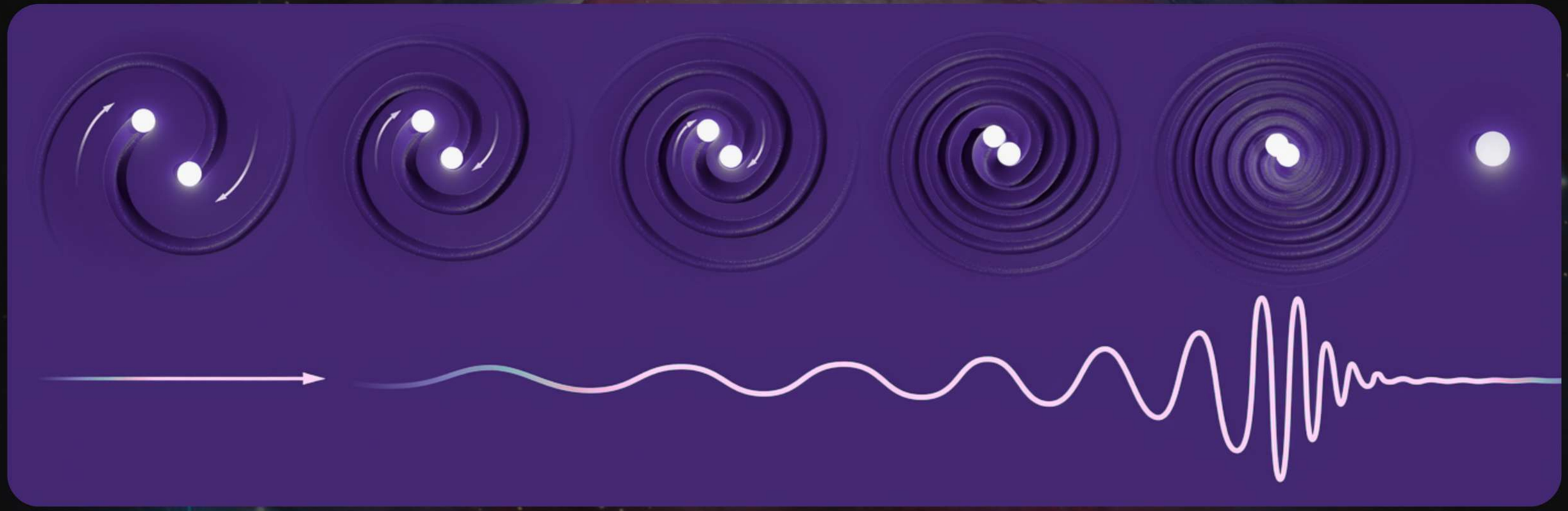
Mass tells spacetime how to curve;
Curved spacetime tells mass how to move.








# GRAVITATIONAL WAVES

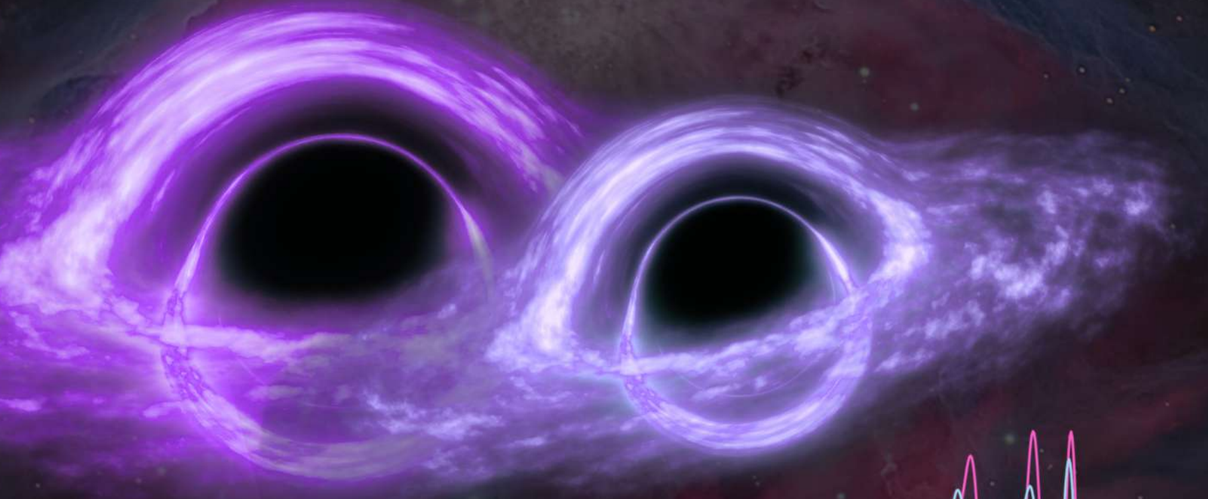


Accelerating massive bodies produce

RIPPLES IN SPACETIME

# THE FIRST DISCOVERY

14 SEPT 2015



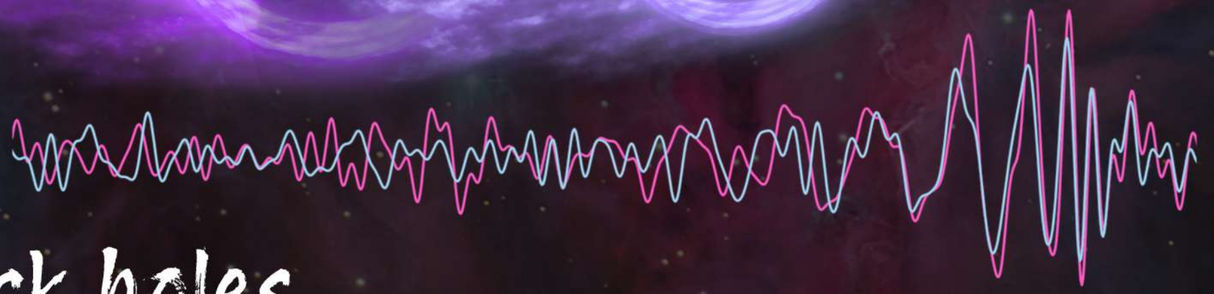
Two merging black holes,
each 30 times the mass of our sun
1.3 billion light years away...

# THE FIRST DISCOVERY

14 SEPT 2015



2017
NOBEL PRIZE
IN PHYSICS






Two merging black holes,
each 30 times the mass of our sun
1.3 billion light years away...

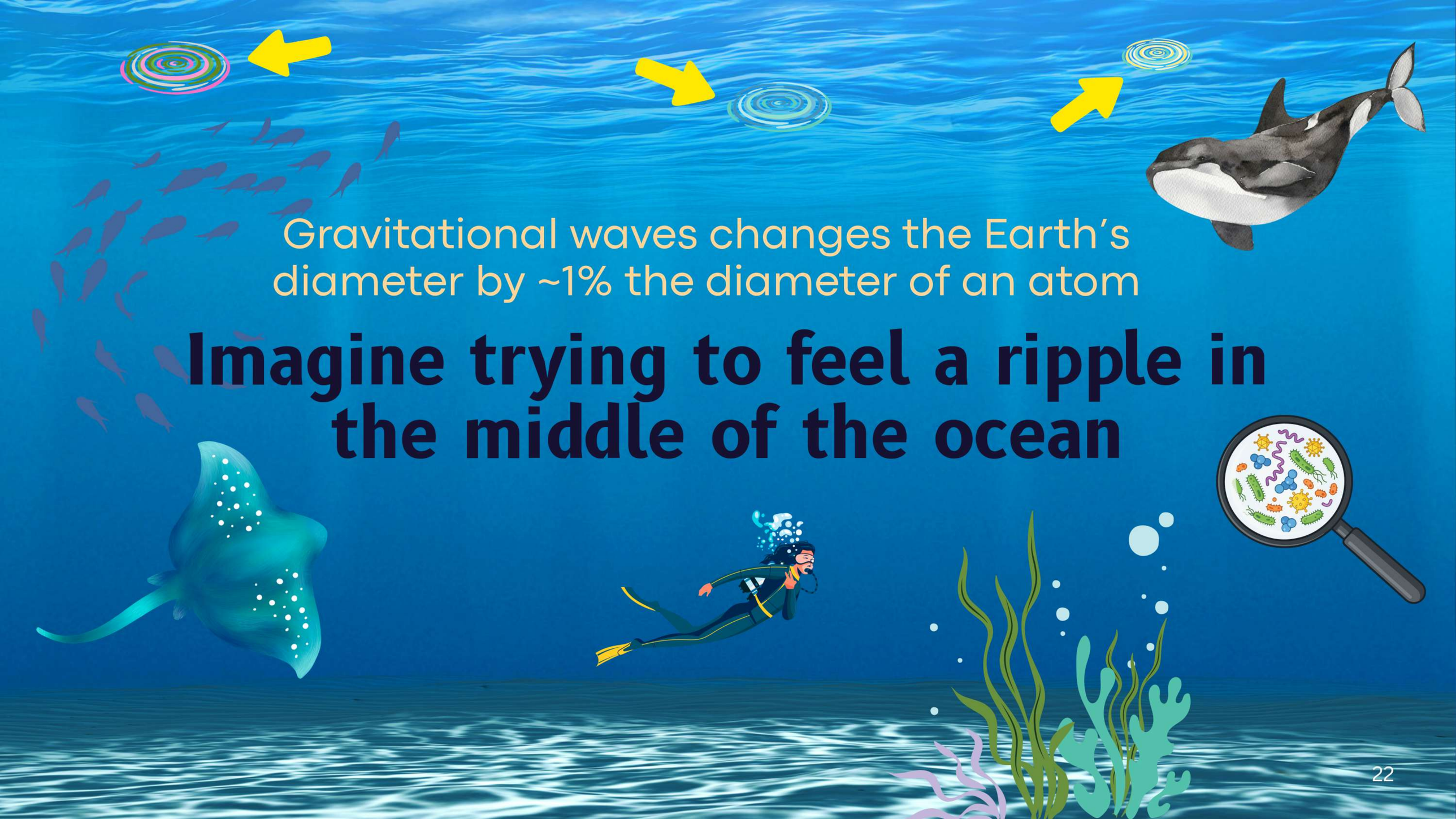
# A NEW ERA OF ASTRONOMY



Gravitational waves propagate unobstructed
across space and time

# CAN YOU FEEL A GRAVITATIONAL WAVE?

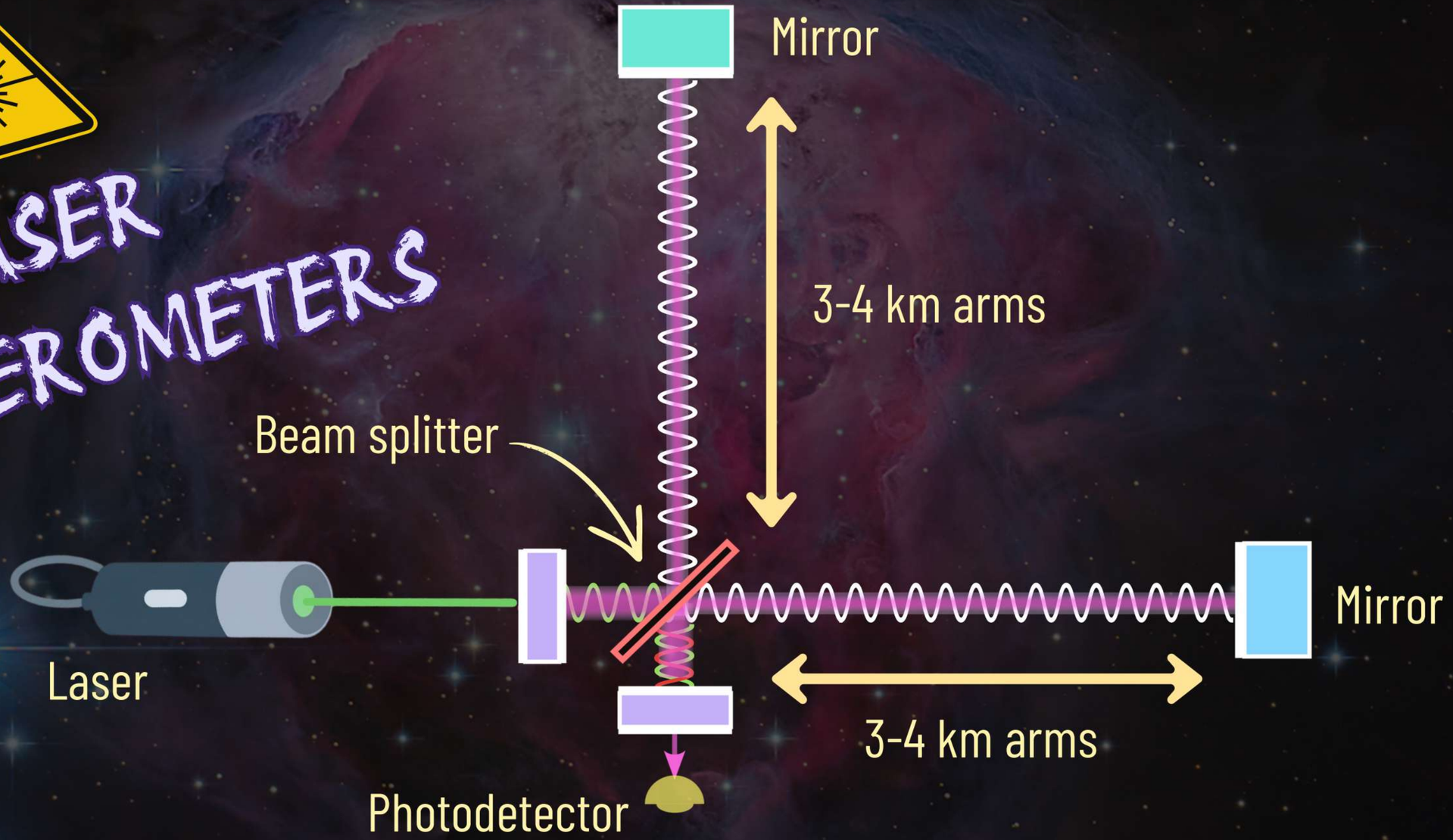


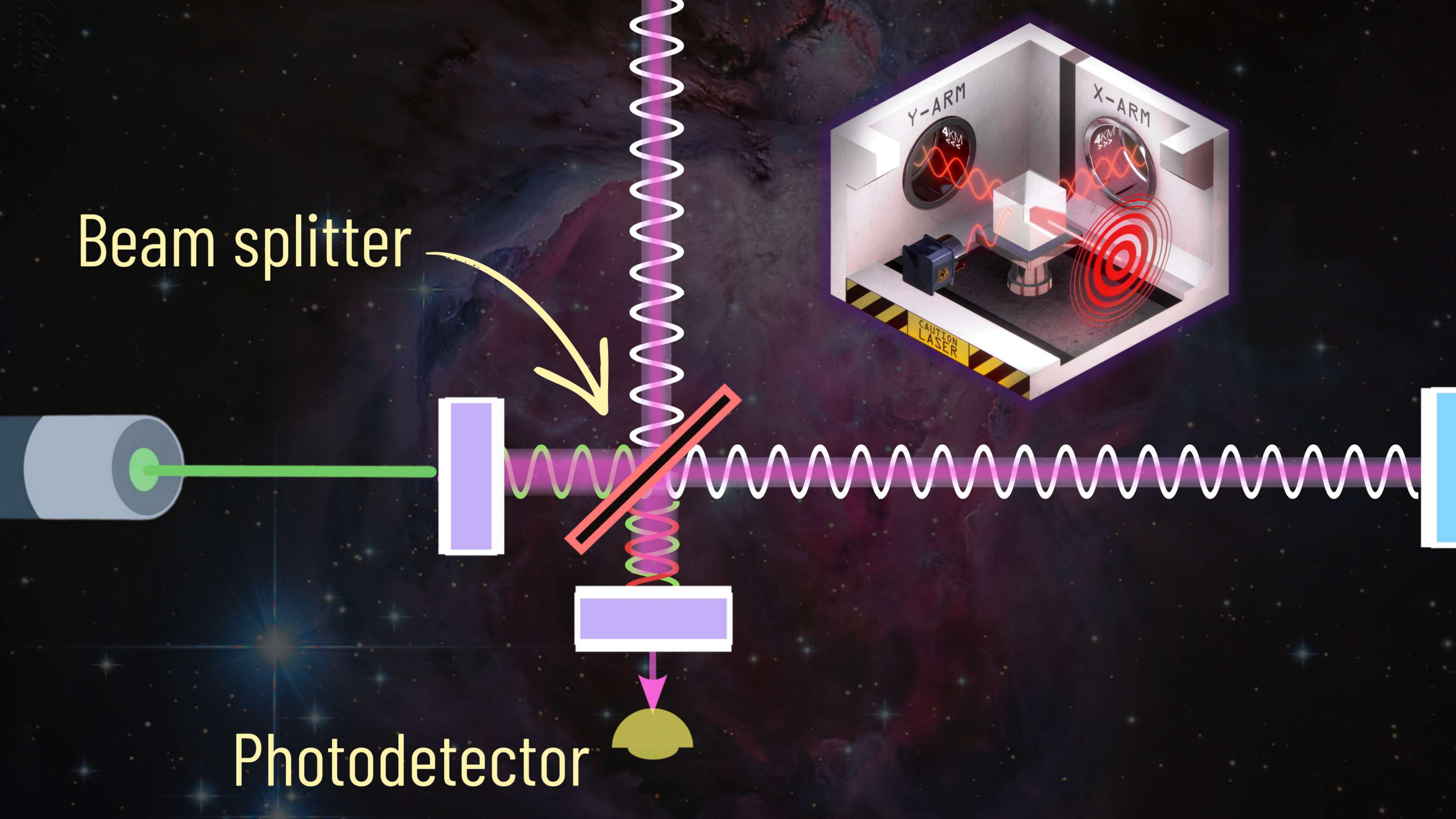

Gravitational waves changes the Earth’s diameter by ~1% the diameter of an atom
Imagine trying to feel a ripple in the middle of the ocean

# HOW DO WE FIND IT?



## LASER INTERFEROMETERS




Beam splitter
Y-ARM
X-ARM
4KM
4KM
CAUTION
LASER
Photodetector

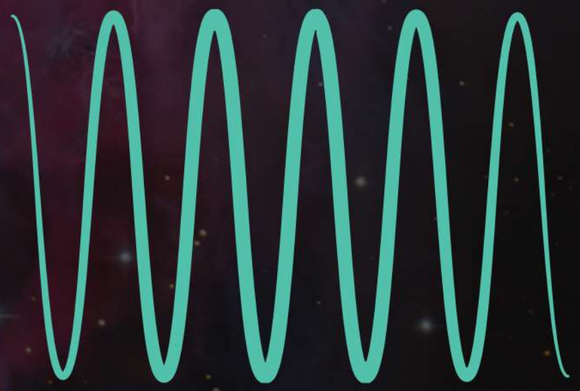
# NO SIGNAL VS DETECTION



Destructive interference
(No signal)



Constructive interference



# GLOBAL NETWORK OF DETECTORS













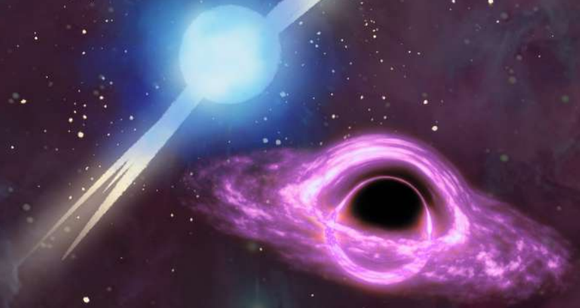
LOADING...

LIGO INDIA

# WHAT HAVE WE "SEEN"?



Black hole mergers



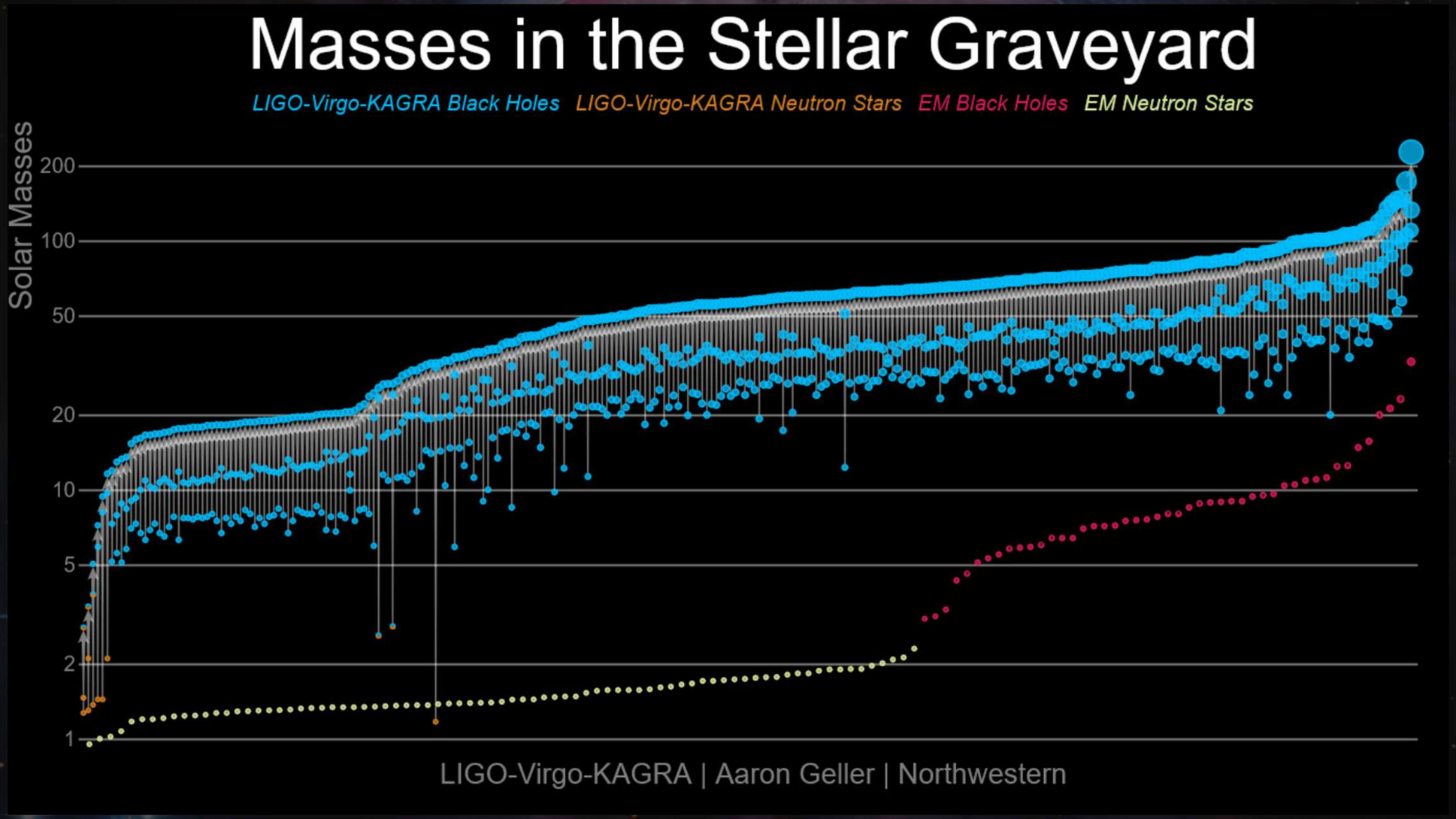
Neutron star-black hole mergers



Neutron star mergers

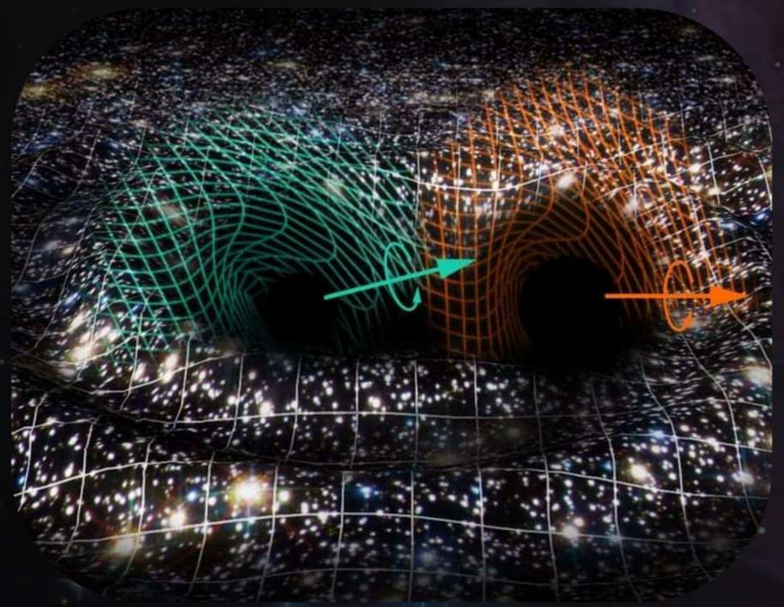
## 390 TOTAL DETECTIONS (!!)

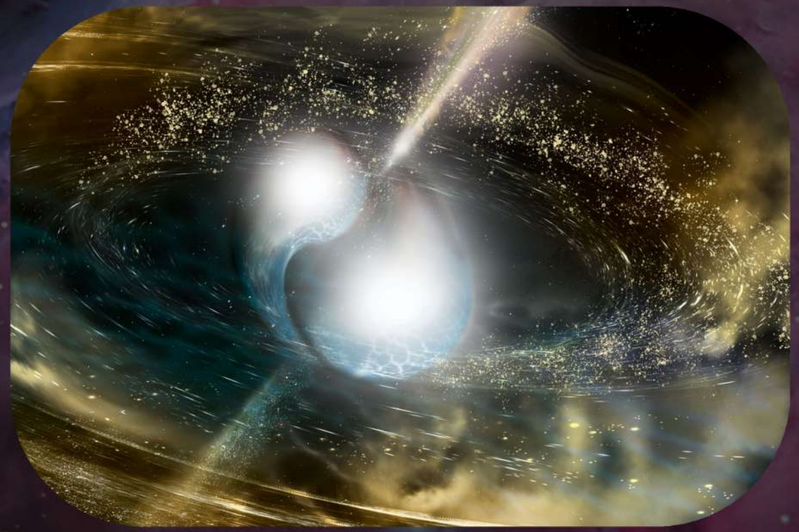
# WHAT HAVE WE "SEEN"?



# MESSAGES FROM DEAD STARS

Testing Einstein's theory of general relativity





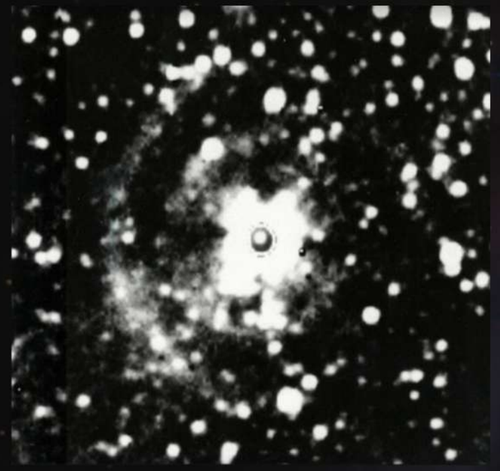
Formation of heavy elements

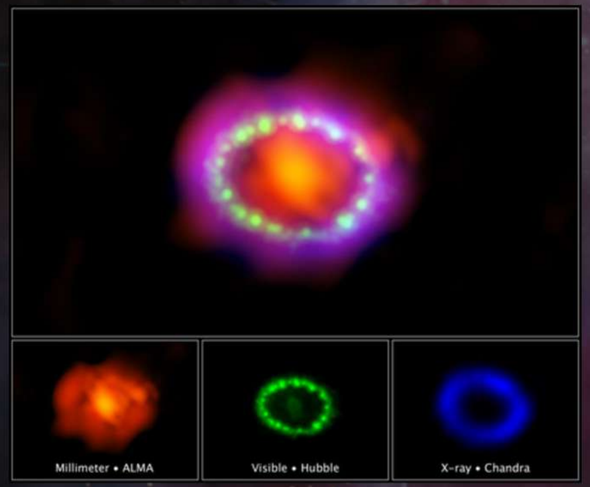



Black hole evolution and population



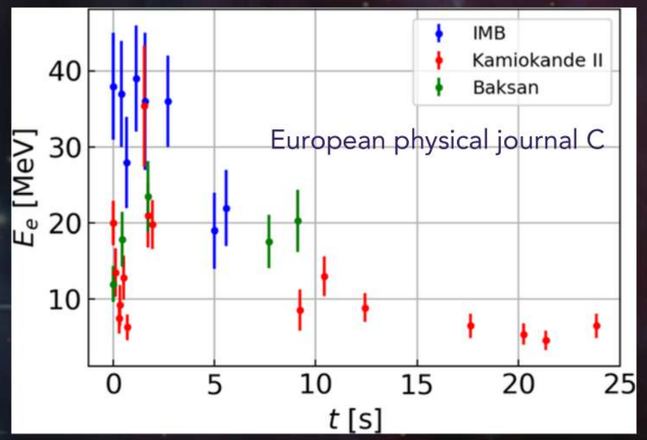
# MULTI-MESSENGER ASTRONOMY





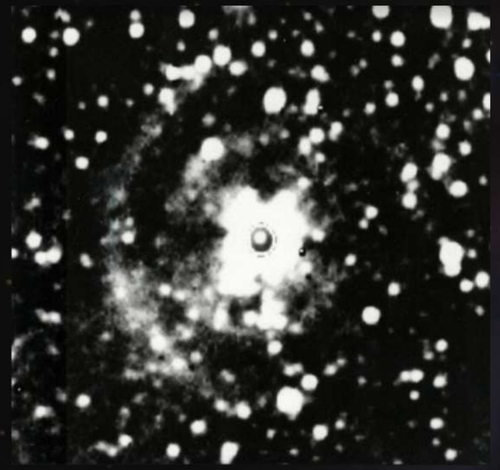


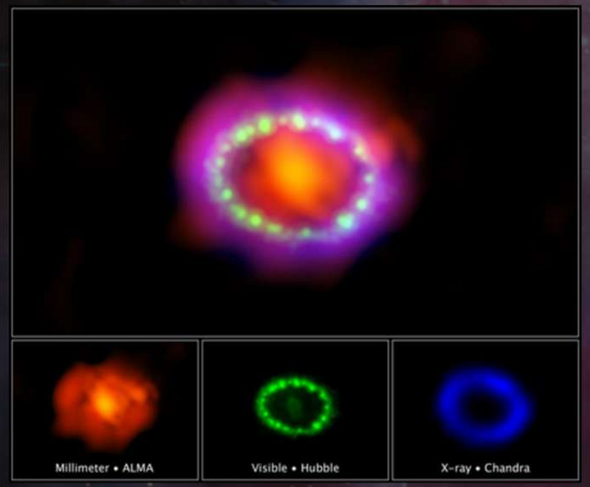





Supernova 1987A

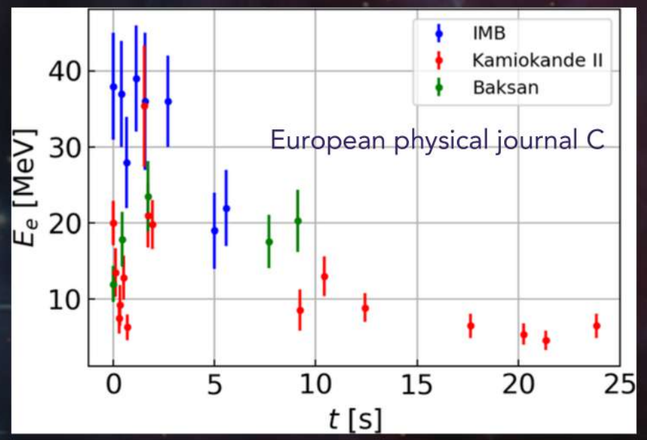
# MULTI-MESSENGER ASTRONOMY







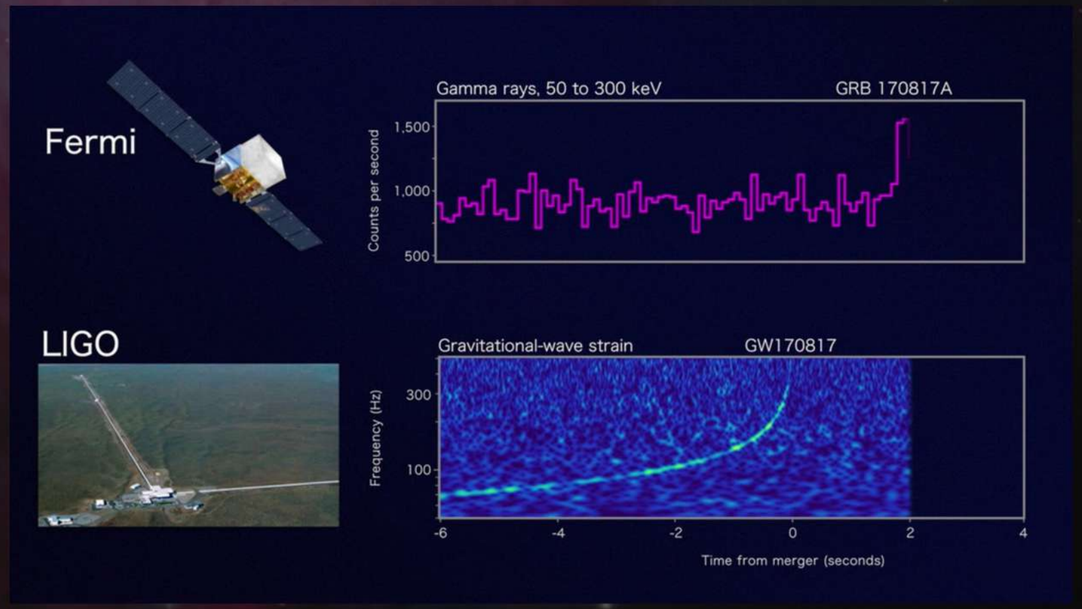





Supernova 1987A






Neutron star merger - 17 Aug 2017

# DESSERT





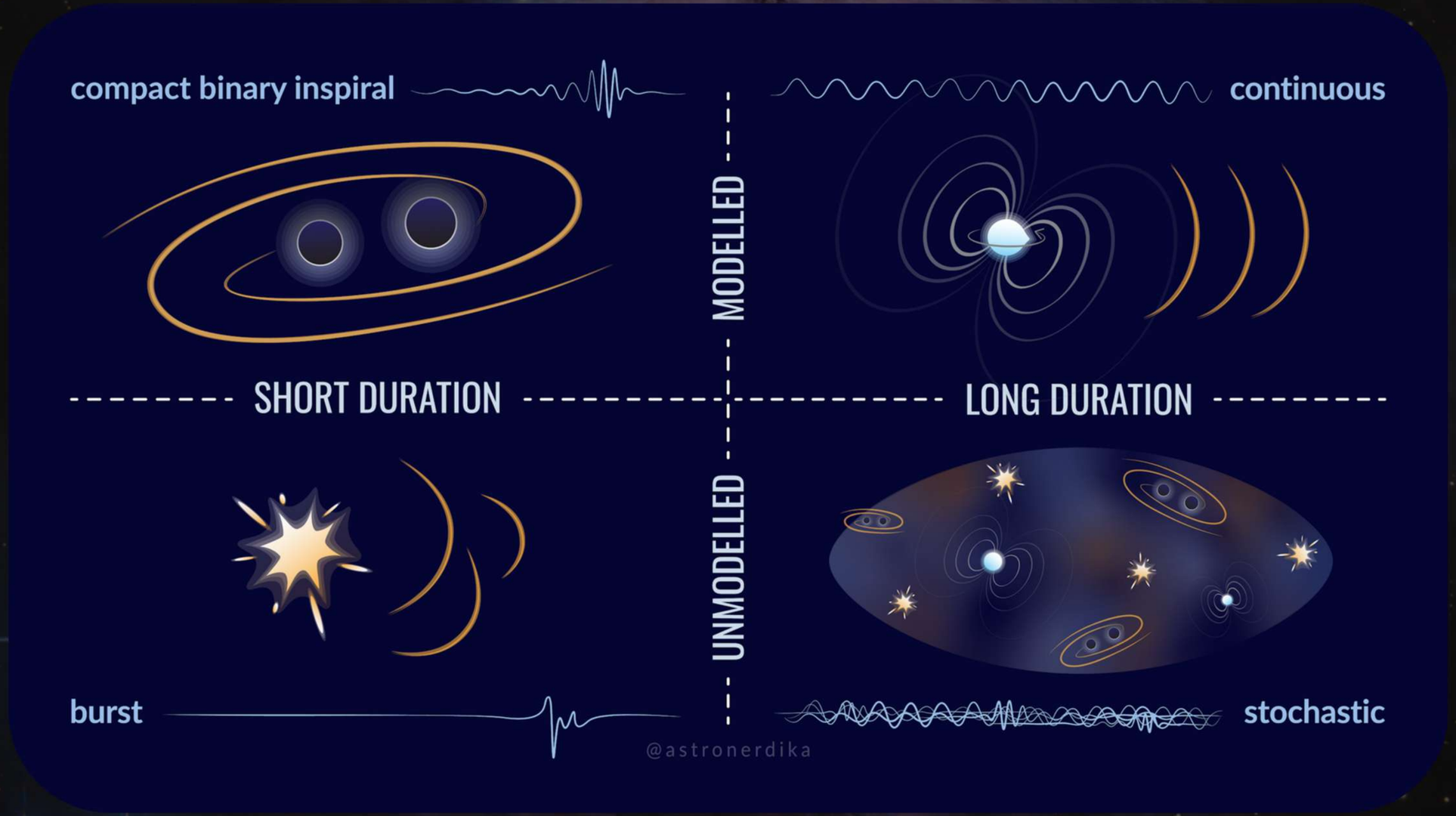


Undetected sources

Future of gravitational-wave astronomy

What do i do?

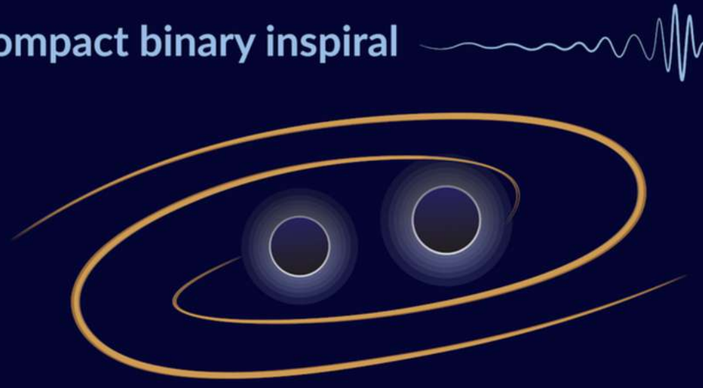
# TYPES OF GRAVITATIONAL WAVES



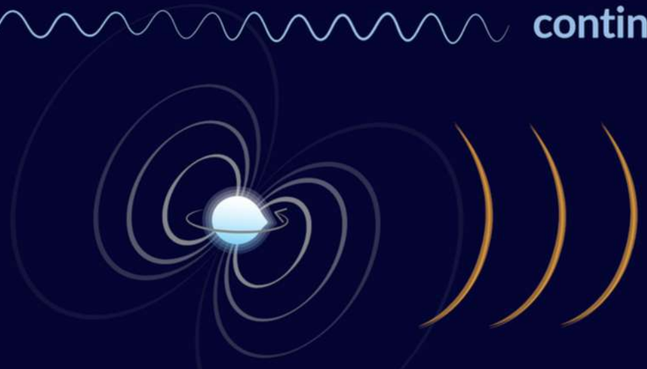
# TYPES OF GRAVITATIONAL WAVES

390 EVENTS DETECTED

compact binary inspiral



continuous



MODELLED

SHORT DURATION

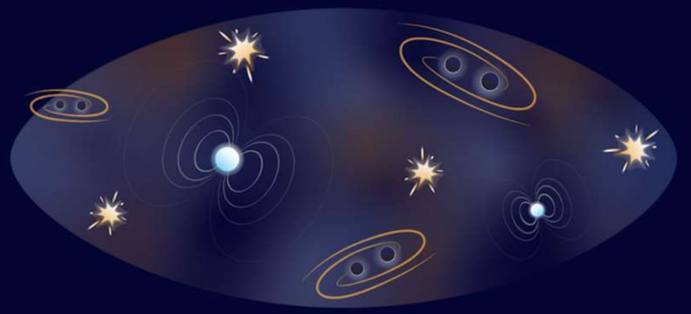
LONG DURATION

UNMODELLED



burst







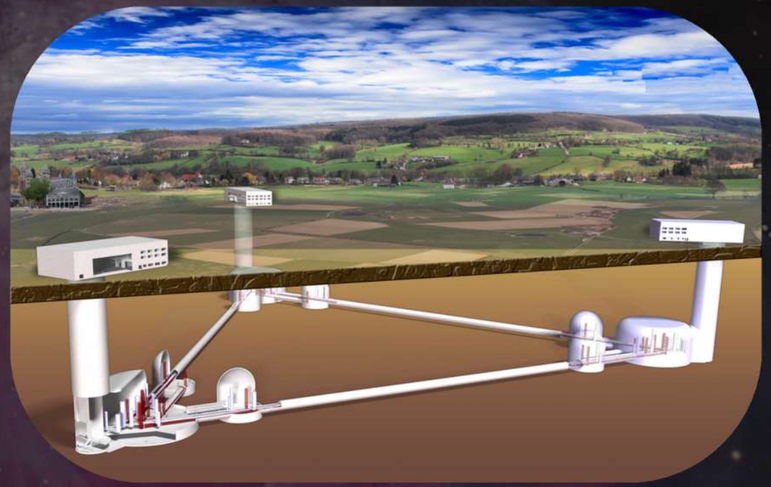
stochastic

@astronerdika

# FUTURE DETECTORS



COSMIC EXPLORER

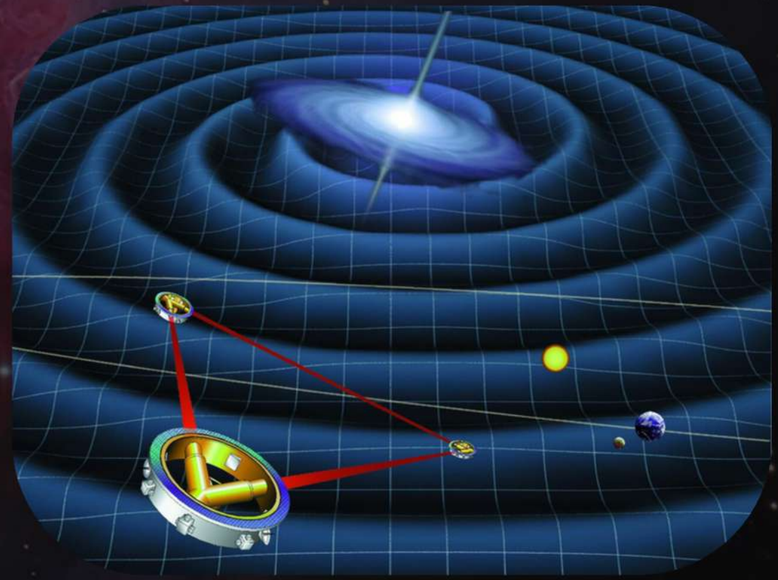


EINSTEIN TELESCOPE

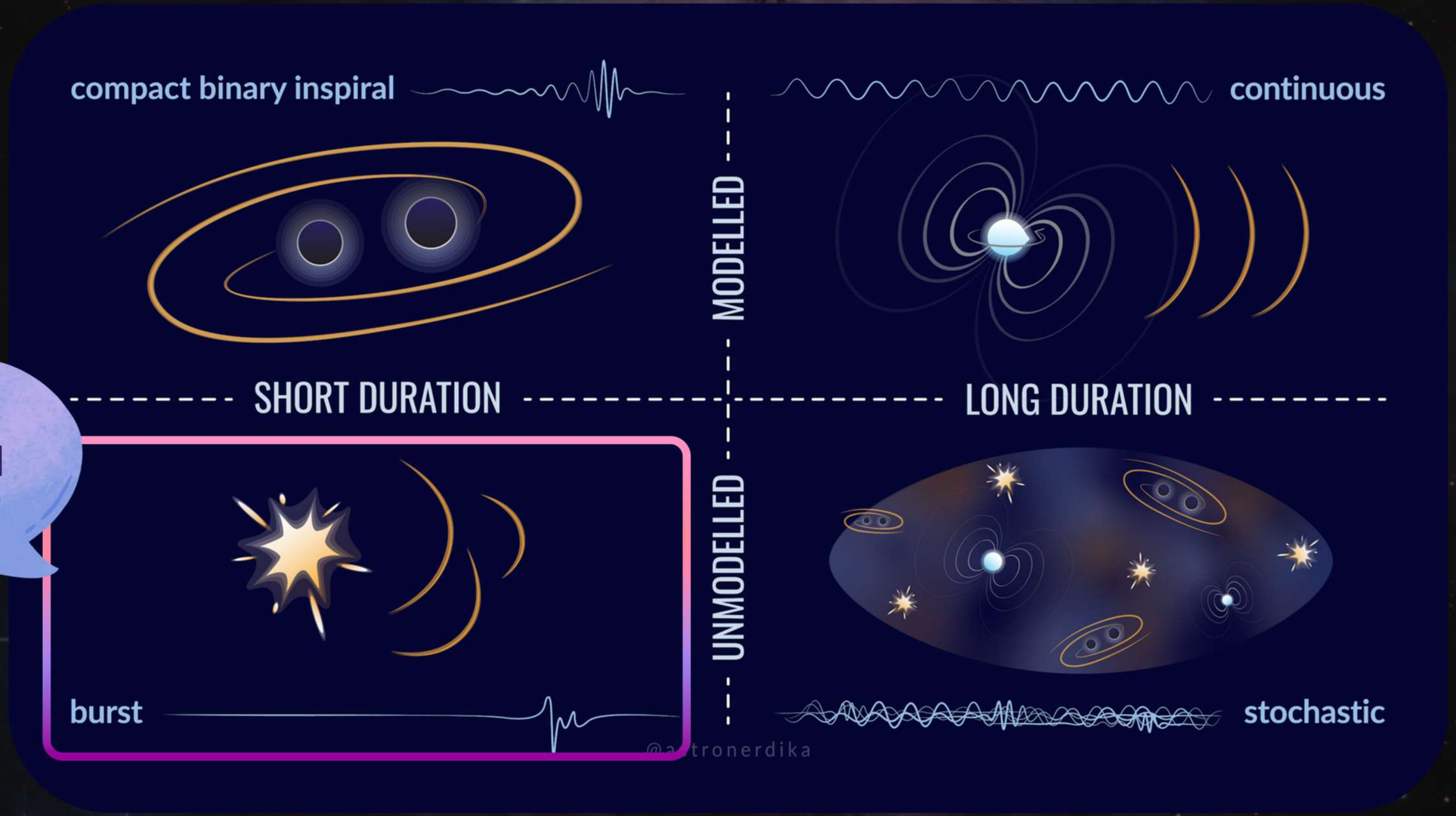
LIGO INDIA



ESA | NASA lisa



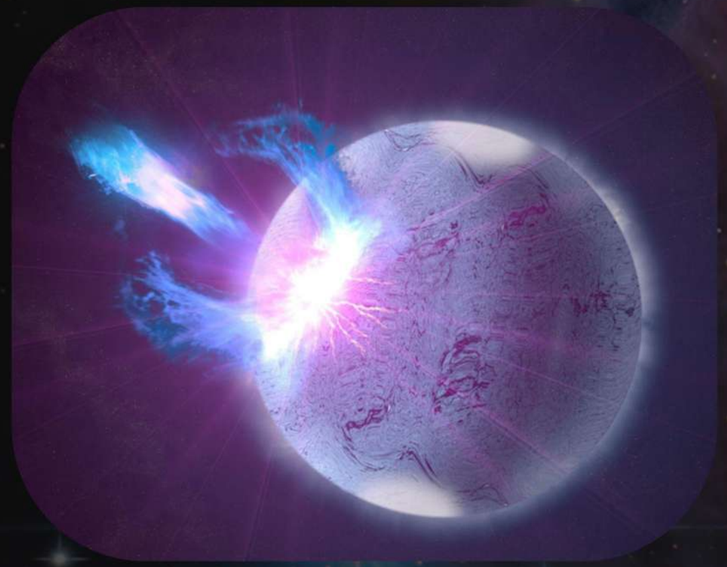
# WHAT DO I DO?




My Research Interest

# FUTURE OF MULTI-MESSENGER

NEUTRON STAR GLITCHES





CORE-COLLAPSE SUPERNOVA

GAMMA RAY BURSTS



"seeing" known astrophysical sources in a different "light"

# READ ABOUT MY RESEARCH




PURSUIT

THE UNIVERSITY OF MELBOURNE

READ MORE

# THE UNKNOWN UNKNOWN



Pontential discovery of new, obscure sources

# TAKE HOME MESSAGE

> GRAVITATIONAL WAVES
> reveal the universe's most violent events,
> from dying stars to cosmic collisions –
> allowing us to explore the cosmos
> through a different lens.












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